

Chapter 2 Introduction to HTML5

Internet & World Wide Web How to Program, 5/e

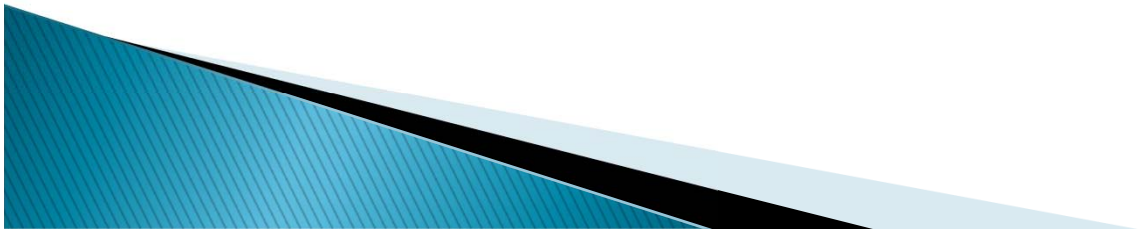
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OBJECTIVES

In this chapter you'll:

- Understand important components of HTML5 documents.
- Use HTML5 to create web pages.
- Add images to web pages.
- Create and use hyperlinks to help users navigate web pages.
- Mark up lists of information.
- Create tables with rows and columns of data.
- Create and use forms to get user input.





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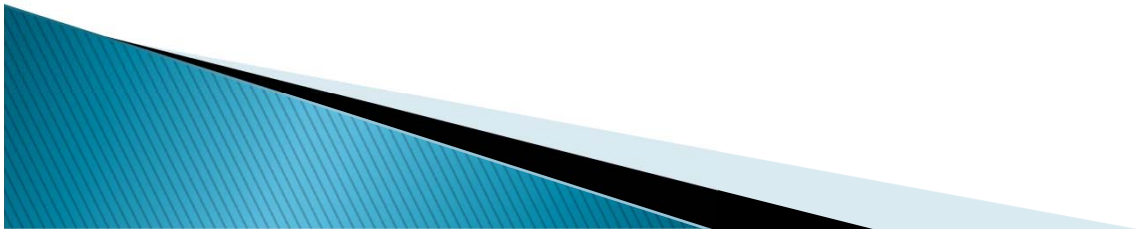
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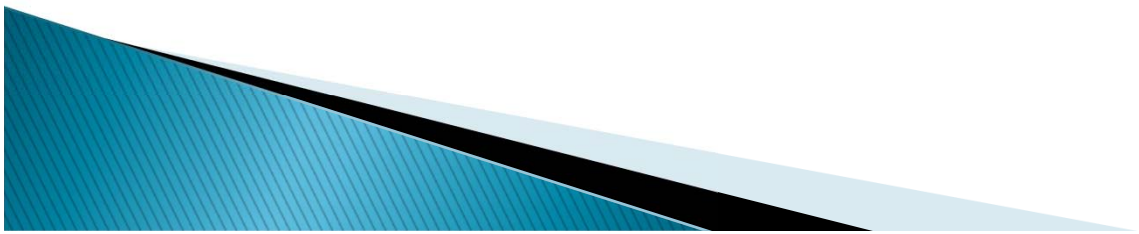
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2.12 Internal Linking

2.13 `meta` Elements

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2.1 Introduction

- HTML5 (HyperText Markup Language 5)
  - markup language that specifies the *structure* and *content* of documents that are displayed in web browsers
- We introduce some basics, then cover more sophisticated HTML5 techniques such as:
  - tables, which are particularly useful for structuring information from databases (i.e., software that stores structured sets of data)
  - forms for collecting information from web-page visitors
  - internal linking for easier page navigation
  - meta elements for specifying information about a document



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2.2 Editing HTML5

- We'll create HTML5 documents by typing HTML5 markup text in a text editor (such as Notepad, TextEdit, vi, emacs) and saving it with the `.html` or `.htm` filename extension.
- Computers called web servers store HTML5 documents.
- Clients (such as web browsers running on your local computer or smartphone) request specific resources such as HTML5 documents from web servers.

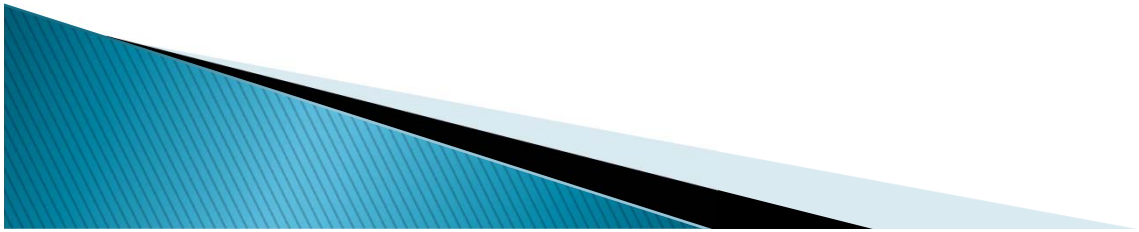


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2.3 First HTML5 Example

- Figure 2.1 is an HTML5 document named `main.html`, which is stored in the `examples/ch02` folder.
- This first example displays the message `Welcome to HTML5!` in the browser.





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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.1: main.html -->
4  <!-- First HTML5 example. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>Welcome</title>
9     </head>
10
11    <body>
12       <p>Welcome to HTML5!</p>
13    </body>
14 </html>
```



Fig. 2.1 | First HTML5 example.



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2.3 First HTML5 Example

Document Type Declaration

- The **document type declaration** (DOCTYPE) is required in HTML5 documents so that browsers render the page in standards mode.
- Some browsers operate in quirks mode to maintain backward compatibility with web pages that are not up-to-date with the latest standards.



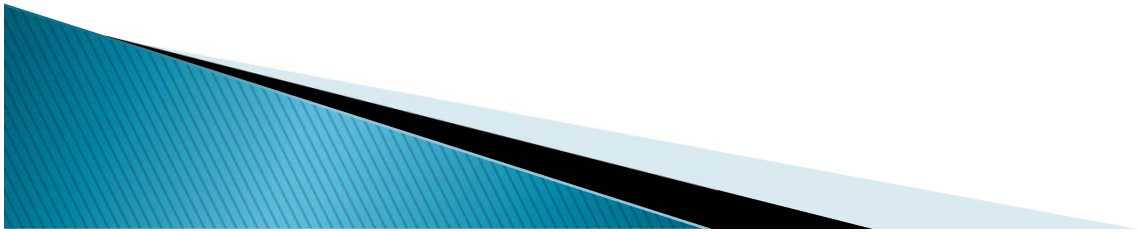
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2.3 First HTML5 Example

Comments

- Insert comments in your HTML5 markup to improve readability and describe the content of a document.
- The browser ignores comments when your document is rendered.
- Comments start with <!-- and end with -->.





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2.3 First HTML5 Example

html, head and body Elements

- HTML5 markup contains text (and images, graphics, animations, audios and videos) that represents the content of a document and elements that specify a document's *structure* and *meaning*.
  - The `html` element *encloses* the head section (represented by the head element) and the body section (represented by the body element).
  - The **head section** contains information about the HTML5 document, such as the character set (UTF-8, the most popular character-encoding scheme for the web) that the page use—which helps the browser determine how to render the content—and the **title**.
  - The head section also can contain special document-formatting instructions called **CSS3 style sheets** and client-side programs called scripts for creating dynamic web pages.
  - The **body section** contains the page's content, which the browser displays when the user visits the web page.

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2.3 First HTML5 Example (cont.)

Start Tags and End Tags

- HTML5 documents *delimit* most elements with a start tag and end tag.
- A **start tag** consists of the element name in angle brackets
  - For example, `<html>`
- An **end tag** consists of the element name preceded by a forward slash (/) in angle brackets
  - For example, `</html>`
- There are several so-called "void elements" that do not have end tags.
- Many start tags have attributes that provide additional information about an element, which browsers use to determine how to process the element.
- Each **attribute** has a **name** and a **value** separated by an equals sign (=).

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Good Programming Practice 2.1

Although HTML5 element and attribute names are case insensitive (you can use uppercase and lowercase letters), it's a good practice to use only lowercase letters.

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2.3 First HTML5 Example (Cont.)

`Title` Element

- The `title` element is called a **nested element**, because it's enclosed in the head element's start and end tags.
- The head element is also a nested element, because it's enclosed in the html element's start and end tags.
- The `title` element describes the web page.
  - Titles usually appear in the title bar at the top of the browser window, in the browser tab on which the page is displayed, and also as the text identifying a page when users add the page to their list of Favorites or Bookmarks, enabling them to return to their favorite sites.
  - Search engines use the title for indexing purposes and when displaying results



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Good Programming Practice 2.2

Indenting nested elements emphasizes a document's structure and promotes readability. We use three spaces for each level of indentation.

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2.3 First HTML5 Example (Cont.)

Paragraph Element (. . .)

- All text placed between the and tags forms one paragraph.



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2.4 W3C HTML5 Validation Service

- HTML5 documents that are syntactically correct are guaranteed to render properly
- HTML5 documents that contain syntax errors may not display properly
- Validation services (e.g., `validator.w3.org/#validate-by-upload`) ensure that an HTML5 document is syntactically correct



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Error-Prevention Tip 2.1

Most browsers attempt to render HTML5 documents even if they're invalid. This can lead to unexpected and undesirable results. Use a validation service, such as the W3C *MarkUp Validation Service*, to confirm that an HTML5 document is syntactically correct.

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2.5 Headings

- HTML5 provides six heading elements (h1 through h6) for specifying the *relative importance* of information
  - Heading element h1 is considered the most significant heading and is rendered in the largest font.
  - Each successive heading element (i.e., h2, h3, etc.) is rendered in a progressively smaller font.



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Portability Tip 2.1

The text size used to display each heading element can vary between browsers. In Chapter 4, we use CSS to control the text size and other text properties.

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Look-and-Feel Observation 2.1

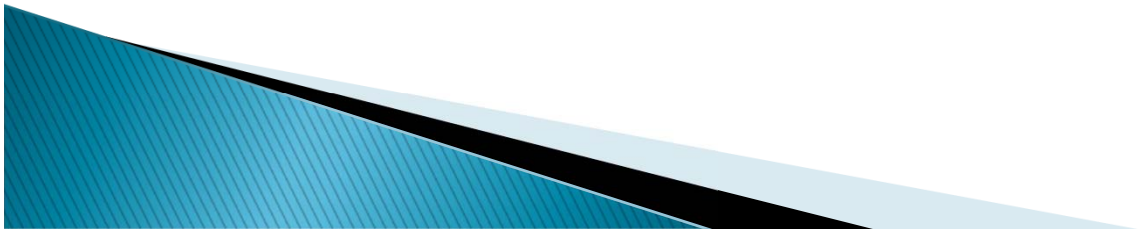
Placing a heading at the top of each page helps viewers understand the purpose of the page. Headers also help create an outline for a document and are indexed by search engines.

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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.2: heading.html -->
4  <!-- Heading elements h1 through h6. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>Headings</title>
9     </head>
10
11    <body>
12       <h1>Level 1 Heading</h1>
13       <h2>Level 2 heading</h2>
14       <h3>Level 3 heading</h3>
15       <h4>Level 4 heading</h4>
16       <h5>Level 5 heading</h5>
17       <h6>Level 6 heading</h6>
18    </body>
19 </html>
```

Fig. 2.2 | Heading elements h1 through h6. (Part 1 of 2.)





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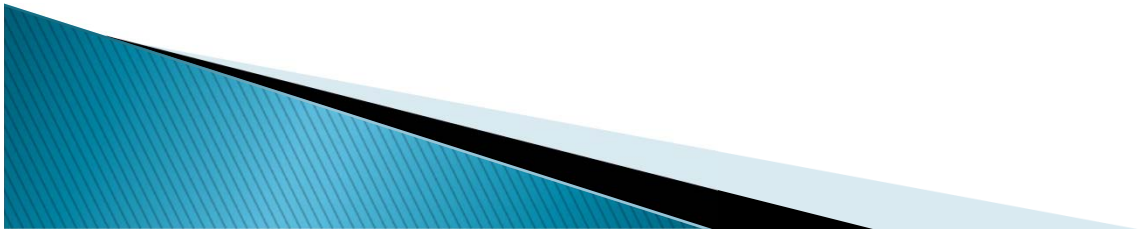
Fig. 2.2 | Heading elements h1 through h6. (Part 2 of 2.)

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2.6 Linking

- A hyperlink references or links to other resources, such as HTML5 documents and images.
- Web browsers typically *underline* text hyperlinks and color them *blue* by default.



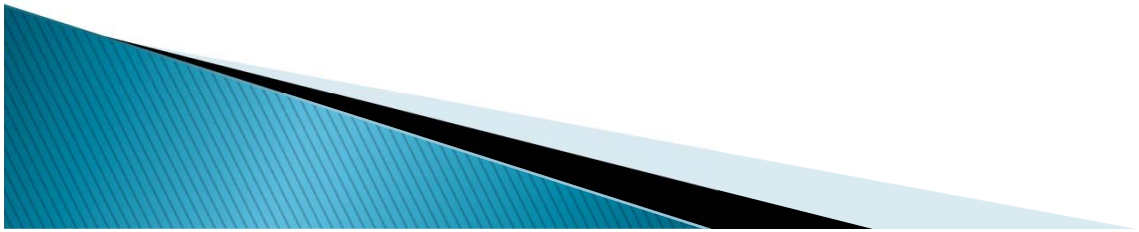


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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.3: links.html -->
4  <!-- Linking to other web pages. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>Links</title>
9     </head>
10
11    <body>
12       <h1>Here are my favorite sites:</h1>
13       <p><strong>Click a name to visit that site.</strong></p>
14
15       <!-- create four text hyperlinks -->
16       <p><a href = "http://www.facebook.com">Facebook</a></p>
17       <p><a href = "http://www.twitter.com">Twitter</a></p>
18       <p><a href = "http://www.foursquare.com">Foursquare</a></p>
19       <p><a href = "http://www.google.com">Google</a></p>
20    </body>
21 </html>
```

Fig. 2.3 | Linking to other web pages. (Part 1 of 2.)





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Fig. 2.3 | Linking to other web pages. (Part 2 of 2.)

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2.6 Linking (Cont.)

- The **`strong` element** indicates that the content has high importance. Browsers typically render such text in a bold font.
- Links are created using the **`a` (`anchor`) element.**
- Attribute **`href` (`hypertext reference`)** specifies a resource's location, such as
  - a web page or location within a web page
  - a file
  - an e-mail address
- When a URL does not indicate a specific document on the website, the web server returns a default web page. This page is often called `index.html`, but most web servers can be configured to use any file as the default web page for the site.
- If the web server cannot locate a requested document, it returns an error indication to the web browser (known as a 404 error), and the browser displays a web page containing an error message.



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Software Engineering Observation 2.1

Although not required in HTML5, enclosing attribute values in either single or double quotes is recommended.

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2.6 Linking (Cont.)

Hyperlinking to an E-Mail Address

- Anchors can link to an e-mail address using a `mailto:` URL
  - When a user clicks this type of anchored link, most browsers launch the default e-mail program (e.g., Mozilla Thunderbird, Microsoft Outlook or Apple Mail) to enable the user to write an e-mail message to the linked address.

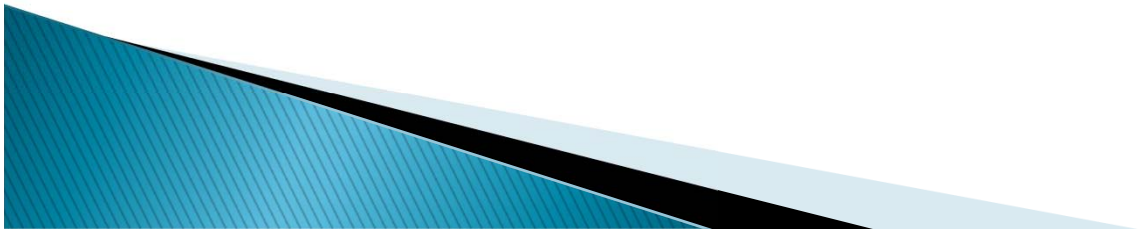


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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.4: contact.html -->
4  <!-- Linking to an e-mail address. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>Contact Page</title>
9     </head>
10
11    <body>
12       <p>
13          To write to <a href = "mailto:deitel@deitel.com">
14          Deitel & Associates, Inc.</a>, click the link and your default
15          email client will open an email message and address it to us.
16       </p>
17    </body>
18 </html>
```

Fig. 2.4 | Linking to an e-mail address. (Part 1 of 3.)





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Fig. 2.4 | Linking to an e-mail address. (Part 2 of 3.)

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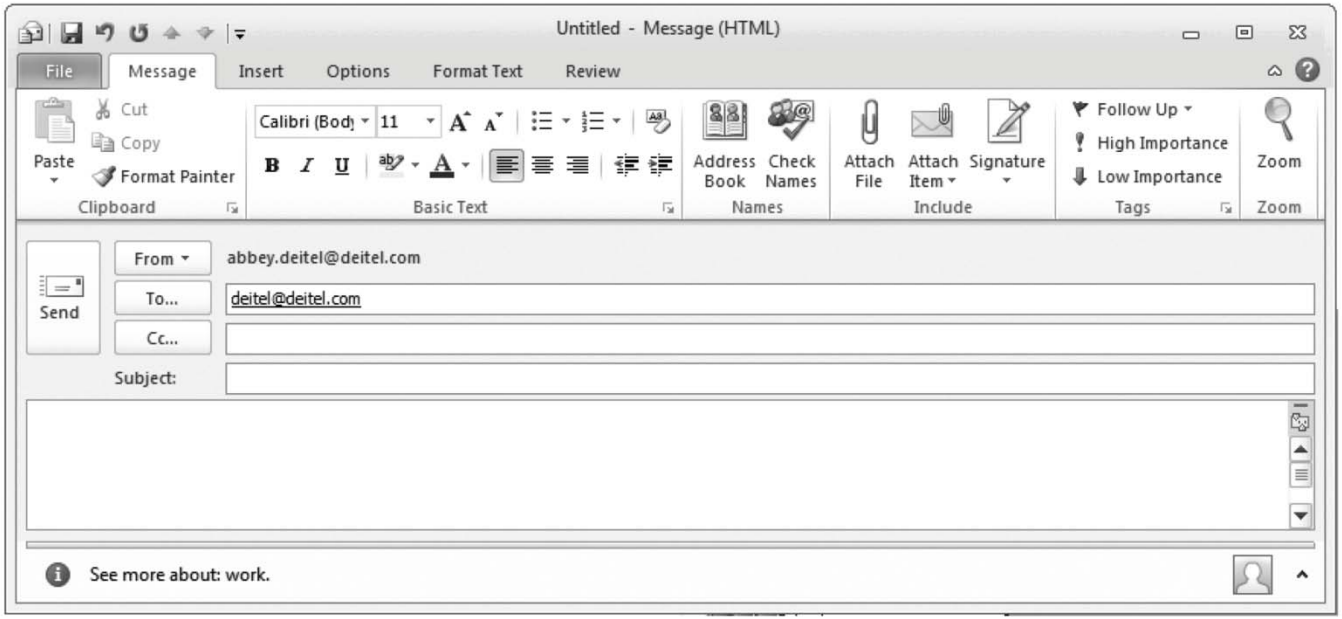


Fig. 2.4 | Linking to an e-mail address. (Part 3 of 3.)

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2.7 Images

- The most popular image formats used by web developers today are PNG (Portable Network Graphics) and JPEG (Joint Photographic Experts Group).
- Users can create images using specialized software, such as Adobe Photoshop Express (www.photoshop.com), G.I.M.P. (www.gimp.org), Inkscape (www.inkscape.org) and many more.
- Images may also be acquired from various websites, many of which offer royalty-free images.

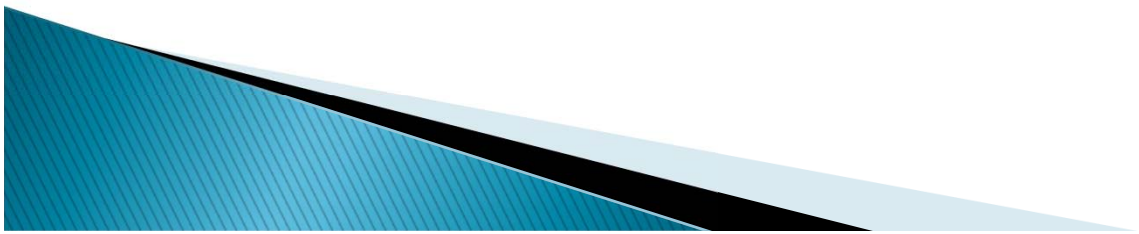


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Image-sharing site	URL
Flickr®	www.flickr.com
Photobucket	photobucket.com
Fotki™	www.fotki.com
deviantART	www.deviantart.com
Picasa™	picasa.google.com
TinyPic®	tinypic.com
ImageShack	www.imageshack.us
FreeDigitalPhotos.net	www.freedigitalphotos.net
Open Stock Photography	www.openstockphotography.org
Open Clip Art Library	www.openclipart.org

Fig. 2.5 | Popular image-sharing sites.



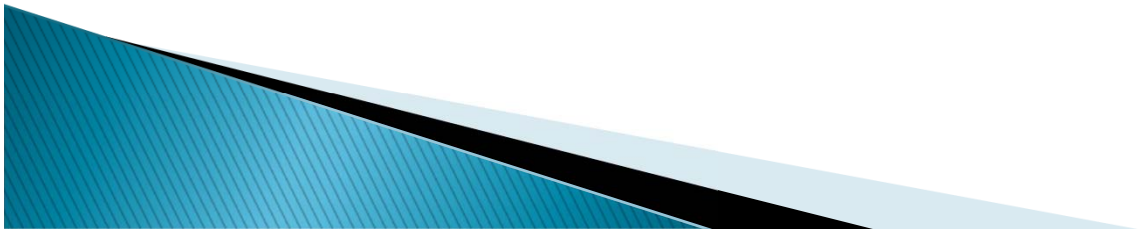


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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.6: picture.html -->
4  <!-- Including images in HTML5 files. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>Images</title>
9     </head>
10
11    <body>
12       <p>
13          <img src = "cpphtp.png" width = "92" height = "120"
14             alt = "C++ How to Program book cover">
15          <img src = "jhtp.png" width = "92" height = "120"
16             alt = "Java How to Program book cover">
17       </p>
18    </body>
19 </html>
```

Fig. 2.6 | Including images in HTML5 files. (Part 1 of 2.)





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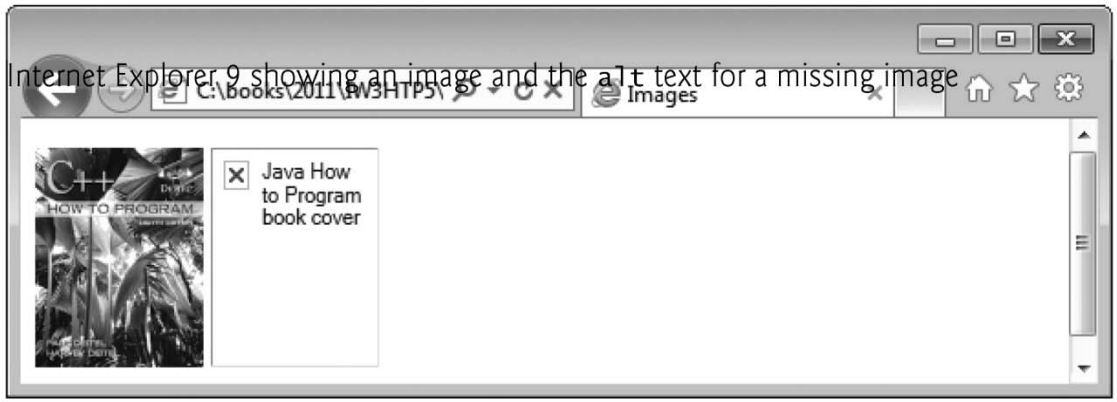


Fig. 2.6 | Including images in HTML5 files. (Part 2 of 2.)

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2.7 Images (cont.)

- The `img` element's `src` attribute specifies an image's location
- Every `img` element must have an `alt` attribute, which contains text that is displayed if the client cannot render the image
  - The `alt` attribute makes web pages more accessible to users with disabilities, especially vision impairments
  - `Width` and `height` are optional attributes
    - If omitted, the browser uses the image's actual width and height
    - Images are measured in pixels



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Performance Tip 2.1

Always include the `width` and the `height` of an image in the `<img>` tag so that when the browser loads the HTML5 file, it will know how much screen space to provide and can lay out the page properly, even before it downloads the image. Including the `width` and `height` attributes in an `<img>` tag can help the browser load and render pages faster.

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Look-and-Feel Observation 2.2

Entering new dimensions for an image that change its width-to-height ratio distorts the appearance of the image. To avoid distortion, if your image is 200 pixels wide and 100 pixels high, for example, any new dimensions should maintain the 2:1 width-to-height ratio.

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2.7 Images (Cont.)

alt Attribute

- A browser may not be able to render an image.
- Every `img` element in an HTML5 document must have an alt attribute.
- If a browser cannot render an image, the browser displays the alt attribute's value.
- The alt attribute is also important for accessibility—speech synthesizer software can speak the alt attribute's value so that a visually impaired user can understand what the browser is displaying. For this reason, the alt attribute should describe the image's contents.



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2.7 Images (Cont.)

Void Elements

- Some HTML5 elements (called **void elements**) contain only attributes and do not mark up text (i.e., text is not placed between a start and an end tag).
- You can terminate void elements (such as the img element) by using the forward slash character (/) inside the closing right angle bracket (>) of the start tag.
- For example, lines 15–16 of Fig. 2.6 could be written as follows:

```
<img src = "jhtp.png" width = "92" height = "120"
   alt = "Java How to Program book cover" />
```



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2.7 Images (Cont.)

Using Images as Hyperlinks

- By using images as hyperlinks, you can create graphical web pages that link to other resources.
- In Fig. 2.7, we create five different image hyperlinks.
- Clicking an image in this example takes the user to a corresponding web page—one of the other examples in this chapter.



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```
<!DOCTYPE html>

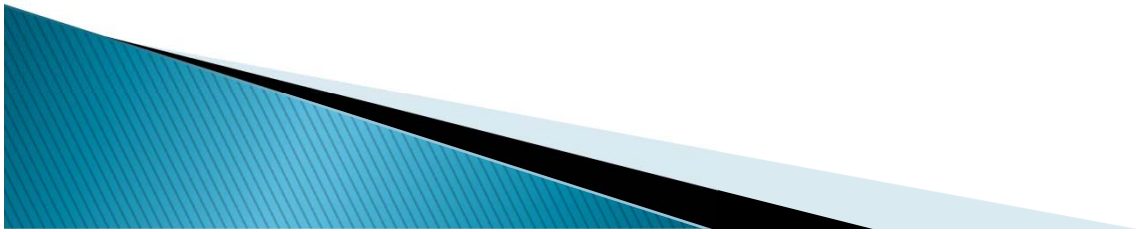
<!-- Fig. 2.7: nav.html -->
<!-- Images as link anchors. -->
<html>
   <head>
      <meta charset = "utf-8">
      <title>Navigation Bar</title>
   </head>

   <body>
      <p>
         <a href = "links.html">
            <img src = "buttons/links.jpg" width = "65"
               height = "50" alt = "Links">
         </a>

         <a href = "list.html">
            <img src = "buttons/list.jpg" width = "65"
               height = "50" alt = "List of Features">
         </a>

```

Fig. 2.7 | Images as link anchors. (Part 1 of 3.)



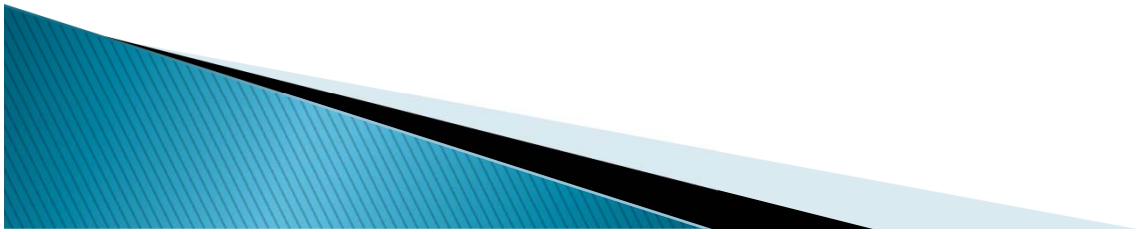


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```
23            <a href = "contact.html">
24               <img src = "buttons/contact.jpg" width = "65"
25                  height = "50" alt = "Contact Me">
26            </a>
27
28            <a href = "table1.html">
29               <img src = "buttons/table.jpg" width = "65"
30                  height = "50" alt = "Tables Page">
31            </a>
32
33            <a href = "form.html">
34               <img src = "buttons/form.jpg" width = "65"
35                  height = "50" alt = "Feedback Form">
36            </a>
37         </p>
38      </body>
39   </html>
```

Fig. 2.7 | Images as link anchors. (Part 2 of 3.)





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Fig. 2.7 | Images as link anchors. (Part 3 of 3.)

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2.8 Special Characters and Horizontal Rules

- HTML5 provides **character entity references** (in the form &*code*;) for representing special characters that cannot be rendered otherwise
- The code can be:
  - Word abbreviations
  - Numbers
    - Decimal
    - Hexadecimal

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Symbol	Description	Character entity reference
HTML5 character entities		
&	ampersand	`&`
'	apostrophe	`'`
>	greater-than	`>`
<	less-than	`<`
"	quote	`"`
Other common character entities		
non-breaking space		` `
©	copyright	`©`
—	em dash	`—`
–	en dash	`–`
¼	fraction 1/4	`¼`
½	fraction 1/2	`½`

Fig. 2.8 | Some common HTML character entity references.

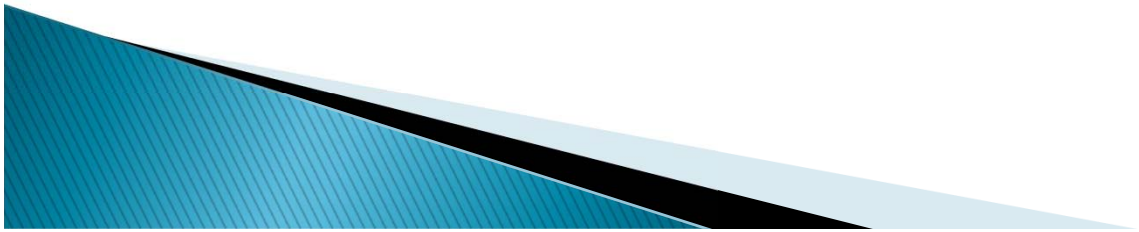


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Symbol	Description	Character entity reference
¾	fraction 3/4	`¾`
…	horizontal ellipsis	`…`
®	registered trademark	`®`
§	section	`§`
™	trademark	`™`

Fig. 2.8 | Some common HTML character entity references.





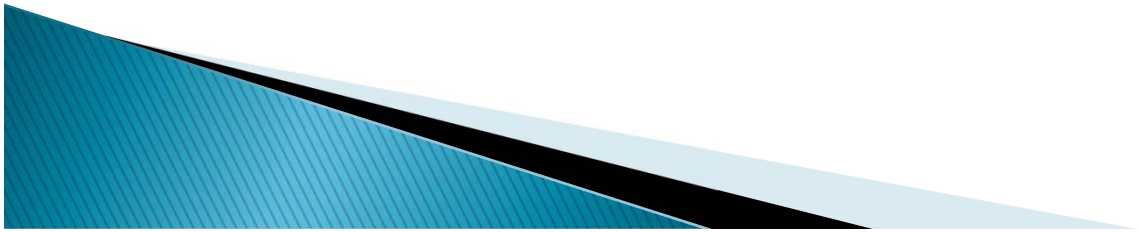
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2.8 Special Characters and Horizontal Rules (Cont.)



- Figure 2.9 demonstrates how to use special characters in an HTML5 document.
- For an extensive list of character entities, see

  www.w3.org/TR/REC-html40/sgml/entities.html



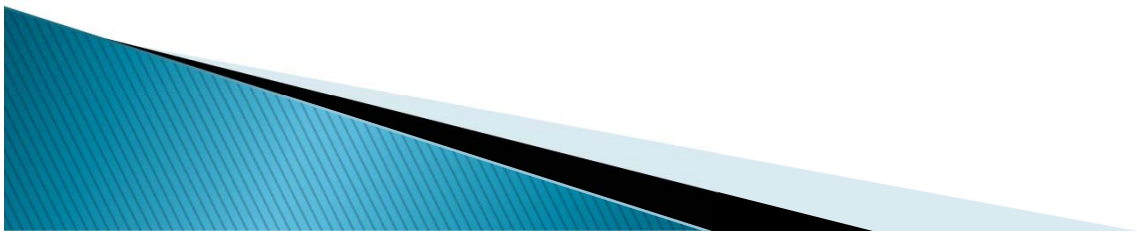


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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.9: contact2.html -->
4  <!-- Inserting special characters. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>Contact Page</title>
9     </head>
10
11    <body>
12       <p>
13          <a href = "mailto:deitel@deitel.com">Send an email to
14          Deitel &amp;amp; Associates, Inc.</a>.
15       </p>
16
17       <hr> <!-- inserts a horizontal rule -->
18
19       <!-- special characters are entered -->
20       <!-- using the form &code; -->
21       <p>All information on this site is <strong>&copy;
22          Deitel & Associates, Inc. 2012.</strong> </p>
23
24       <!-- to strike through text use <del> element -->
```

Fig. 2.9 | Inserting special characters. (Part 1 of 2.)





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```
25          <!-- to subscript text use <sub> element -->
26          <!-- to superscript text use <sup> element -->
27          <!-- these elements are nested inside other elements -->
28          <p><del>You may download 3.14 x 10<sup>2</sup>
29             characters worth of information from this site.</del>
30             The first item in the series is x<sub>1</sub>.</p>
31          <p>Note: &lt; &frac14; of the information
32             presented here is updated daily.</p>
33       </body>
34    </html>
```

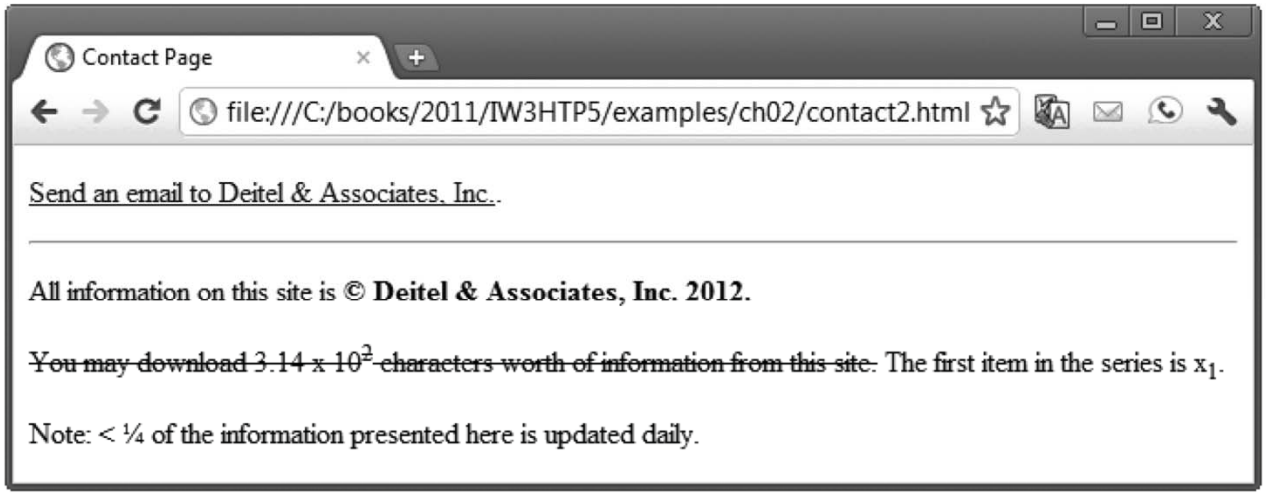


Fig. 2.9 | Inserting special characters. (Part 2 of 2.)

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2.8 Special Characters and Horizontal Rules (Cont.)



- A **horizontal rule**, indicated by the **<hr>** tag renders a horizontal line with extra space above and below it in most browsers.
- The horizontal rule element should be considered a legacy element and you should avoid using it.
- CSS can be used to add horizontal rules and other formatting to documents.
- Special characters can also be represented as **numeric character references**—decimal or hexadecimal (hex) values representing special characters.
  - For example, the & character is represented in decimal and hexadecimal notation as & and &, respectively.
- Hexadecimal numbers are discussed in Appendix E, Number Systems, which is available online at www.deitel.com/books/iw3htp5/.



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2.9 Lists

- Unordered list element `ul`
  - creates a list in which each item in the list begins with a bullet symbol (typically a disc)
  - Each entry is an `li` (list item) element. Most web browsers render these elements with a line break and a bullet symbol at the beginning of the line.



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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.10: links2.html -->
4  <!-- Unordered list containing hyperlinks. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>Links</title>
9     </head>
10
11    <body>
12       <h1>Here are my favorite sites</h1>
13       <p><strong>Click on a name to go to that page</strong></p>
14
15       <!-- create an unordered list -->
16       <ul>
17          <!-- the list contains four list items -->
18          <li><a href = "http://www.youtube.com">YouTube</a></li>
19          <li><a href = "http://www.wikipedia.org">Wikipedia</a></li>
20          <li><a href = "http://www.amazon.com">Amazon</a></li>
21          <li><a href = "http://www.linkedin.com">LinkedIn</a></li>
22       </ul>
23    </body>
24 </html>
```

Fig. 2.10 | Unordered list containing hyperlinks. (Part 1 of 2.)



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Fig. 2.10 | Unordered list containing hyperlinks. (Part 2 of 2.)

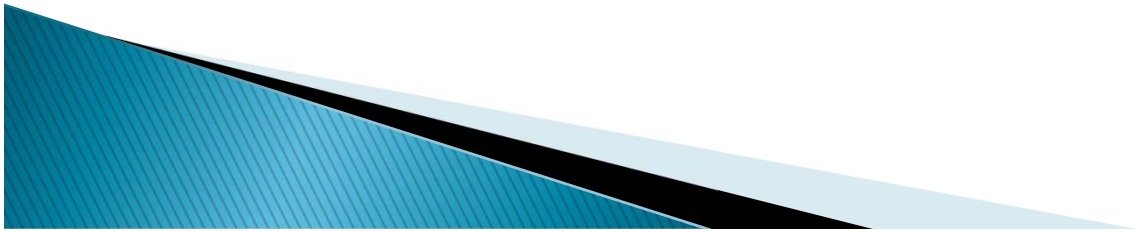
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2.9 Lists (Cont.)

Nested Lists

- Lists may be *nested* to represent *hierarchical* relationships, as in a multi-level outline.
- Figure 2.11 demonstrates nested lists and ordered lists.
- The ordered-list element `ol` creates a list in which each item begins with a number.



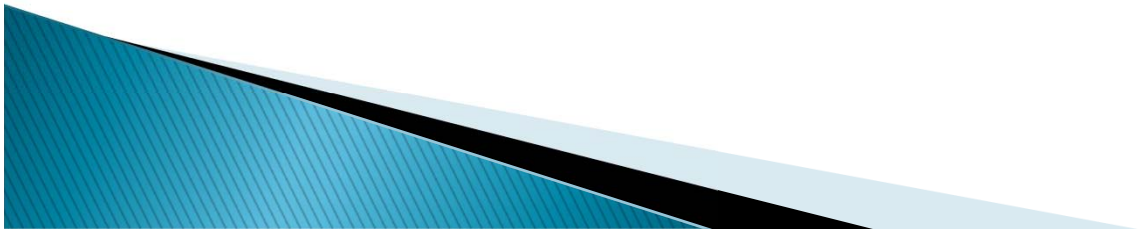


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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.11: list.html -->
4  <!-- Nested lists and ordered lists. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>Lists</title>
9     </head>
10
11    <body>
12       <h1>The Best Features of the Internet</h1>
13
```

Fig. 2.11 | Nested lists and ordered lists. (Part 1 of 4.)



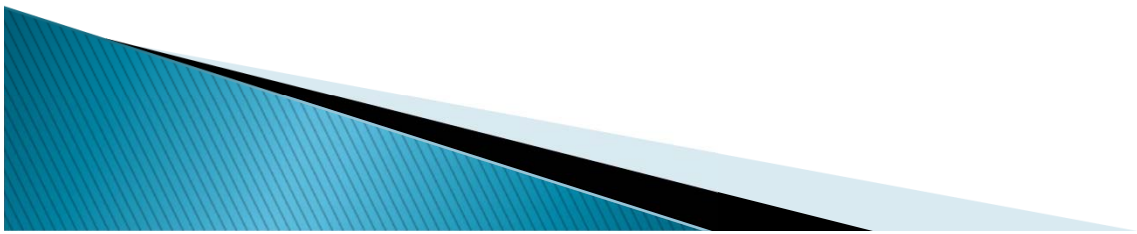


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```
14          <!-- create an unordered list -->
15          <ul>
16             <li>You can meet new people from countries around
17                the world.</li>
18             <li>
19                You have access to new media as it becomes public:
20
21                <!-- this starts a nested unordered list, which uses a -->
22                <!-- different bullet. The list ends when you -->
23                <!-- close the <ul> tag. -->
24                <ul>
25                   <li>New games</li>
26                   <li>New applications
27
28                      <!-- nested ordered list -->
29                      <ol>
30                         <li>For business</li>
31                         <li>For pleasure</li>
32                      </ol>
33                   </li> <!-- ends line 27 new applications li-->
34
35                   <li>Around the clock news</li>
36                   <li>Search engines</li>
37                   <li>Shopping</li>
```

Fig. 2.11 | Nested lists and ordered lists. (Part 2 of 4.)



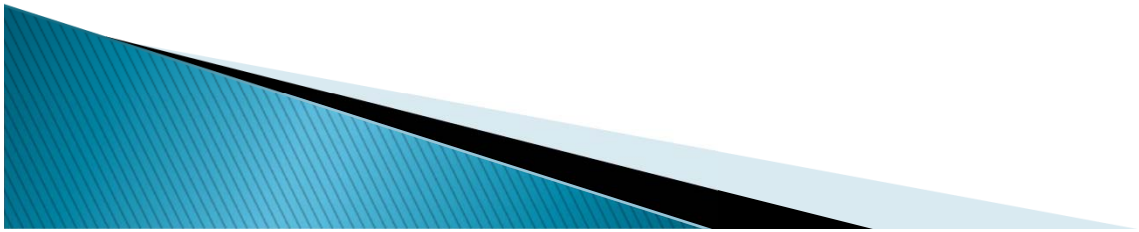


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```
38                <li>Programming
39
40                   <!-- another nested ordered list -->
41                   <ol>
42                      <li>XML</li>
43                      <li>Java</li>
44                      <li>HTML5</li>
45                      <li>JavaScript</li>
46                      <li>New languages</li>
47                   </ol>
48                </li> <!-- ends programming li of line 38 -->
49             </ul> <!-- ends the nested list of line 24 -->
50          </li>
51
52          <li>Links</li>
53          <li>Keeping in touch with old friends</li>
54          <li>It's the technology of the future!</li>
55       </ul> <!-- ends the unordered list of line 15 -->
56    </body>
57 </html>
```

Fig. 2.11 | Nested lists and ordered lists. (Part 3 of 4.)





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Fig. 2.11 | Nested lists and ordered lists. (Part 4 of 4.)

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2.10 Tables

- Tables are frequently used to organize data into *rows* and *columns*.
- The `table` element defines an HTML5 table
- The `summary` attribute summarizes the table's contents and is used by speech devices to make the table more accessible to users with visual impairments.
- The `caption` element specifies a table's title.
- It's good practice to include a general description of a table's information in the table element's **`summary`** attribute—one of the many HTML5 features that make web pages more accessible to users with disabilities.
  - Speech devices use this attribute to make the table more accessible to users with visual impairments.

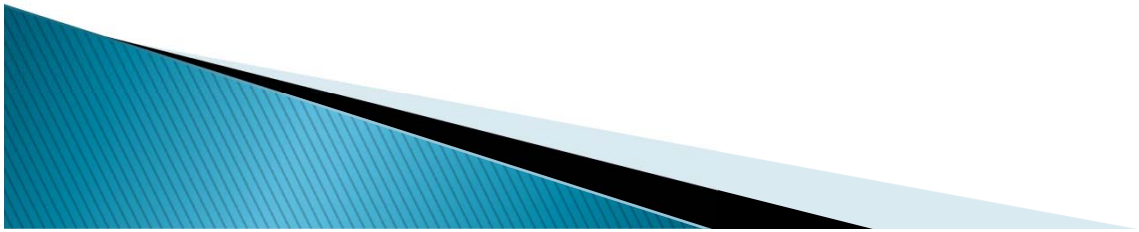


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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.12: table1.html -->
4  <!-- Creating a basic table. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>A simple HTML5 table</title>
9     </head>
10
11    <body>
12       <!-- the <table> tag opens a table -->
13       <table border = "1">
14
15          <!-- the <caption> tag summarizes the table's -->
16          <!-- contents (this helps visually impaired people) -->
17          <caption><strong>Table of Fruits (1st column) and
18             Their Prices (2nd column)</strong></caption>
19
```

Fig. 2.12 | Creating a basic table. (Part 1 of 4.)





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```
20          <!-- the <thead> section appears first in the table -->
21          <!-- it formats the table header area -->
22          <thead>
23             <tr> <!-- <tr> inserts a table row -->
24                <th>Fruit</th> <!-- insert a heading cell -->
25                <th>Price</th>
26             </tr>
27          </thead>
28
29          <!-- the <tfoot> section appears last in the table -->
30          <!-- it formats the table footer -->
31          <tfoot>
32             <tr>
33                <th>Total</th>
34                <th>$3.75</th>
35             </tr>
36          </tfoot>
37
38          <!-- all table content is enclosed -->
39          <!-- within the <tbody> -->
40          <tbody>
41             <tr>
42                <td>Apple</td> <!-- insert a data cell -->
43                <td>$0.25</td>
44             </tr>
```

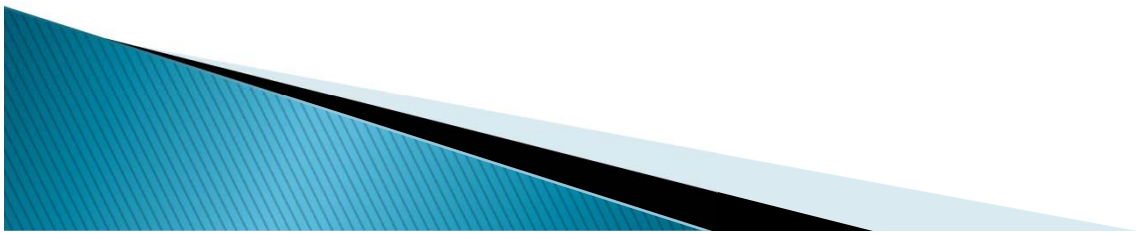
Fig. 2.12 | Creating a basic table. (Part 2 of 4.)



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```
45                <tr>
46                   <td>Orange</td>
47                   <td>$0.50</td>
48                </tr>
49                <tr>
50                   <td>Banana</td>
51                   <td>$1.00</td>
52                </tr>
53                <tr>
54                   <td>Pineapple</td>
55                   <td>$2.00</td>
56                </tr>
57             </tbody>
58          </table>
59       </body>
60    </html>
```

Fig. 2.12 | Creating a basic table. (Part 3 of 4.)



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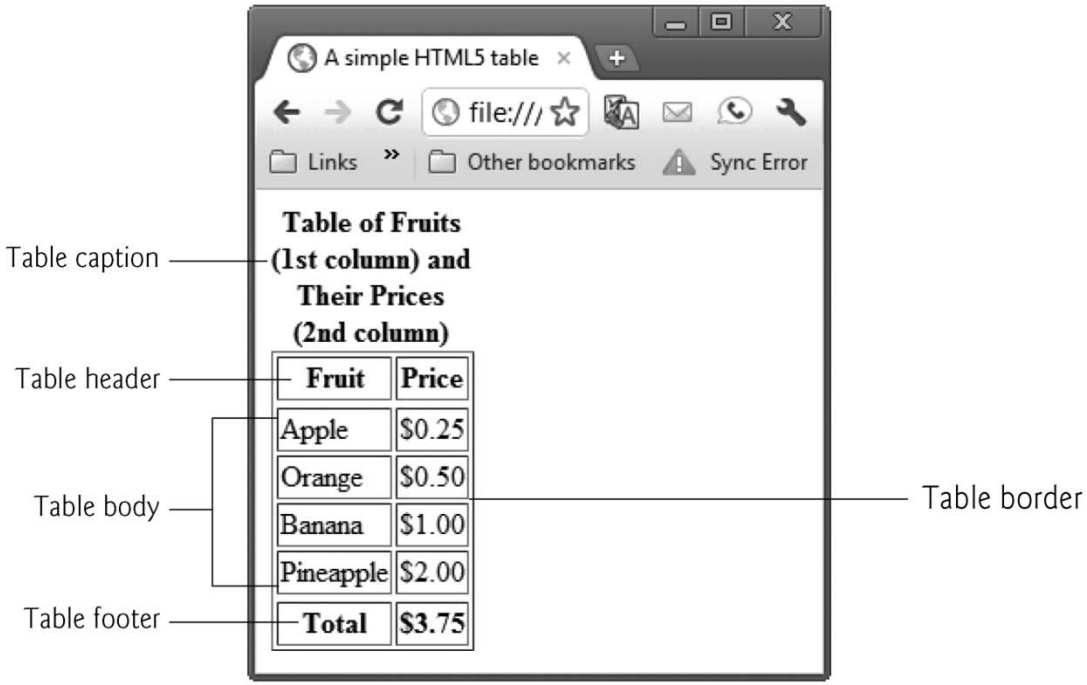


Fig. 2.12 | Creating a basic table. (Part 4 of 4.)

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2.10 Tables (Cont.)

- A table can be split into three distinct sections:
  - Head (`thead` element)
    - Table titles
    - Column headers
  - Body (`tbody` element)
    - Primary table data
  - Table Foot (`tfoot` element)
    - Calculation results
    - Footnotes
    - Above body section in the code, but displays at the bottom in the page

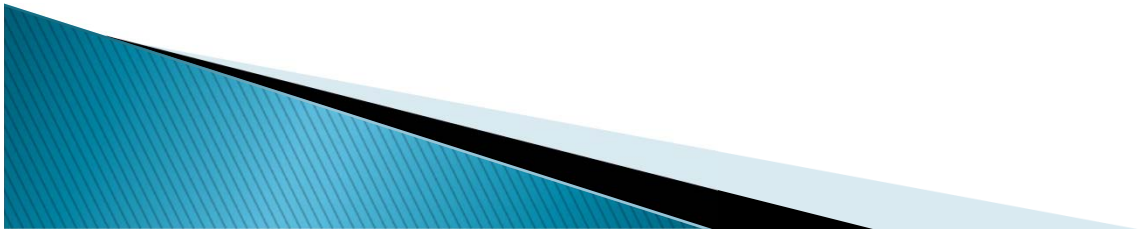


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2.10 Tables (Cont.)

- `tr` Element
  - Defines individual table rows
  - Element `th`
    - Defines a header cell
- `Td` Element
  - Contains table data elements





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2.10 Tables (Cont.)

Using rowspan and colspan with Tables

- Figure 2.13 introduces two new attributes that allow you to build more complex tables.
- You can merge data cells with the `rowspan` and `colspan` attributes
  - The values of these attributes specify the number of rows or columns occupied by the cell.
  - Can be placed inside any data cell or table header cell.
- The `br` element is render as a line break in most browsers—any markup or text following a `br` element is rendered on the next line.
- Like the `img` element, `br` is an example of a void element.
- Like the `hr` element, `br` is considered a legacy formatting element that you should avoid using—in general, formatting should be specified using CSS.

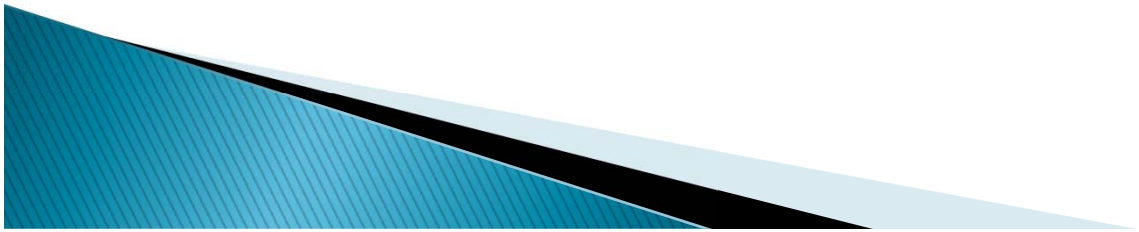


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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.13: table2.html -->
4  <!-- Complex HTML5 table. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>Tables</title>
9     </head>
10
11    <body>
12       <h1>Table Example: Spanning Rows and Columns</h1>
13
14       <table border = "1">
15          <caption>A more complex sample table</caption>
16
```

Fig. 2.13 | Complex HTML5 table. (Part 1 of 4.)



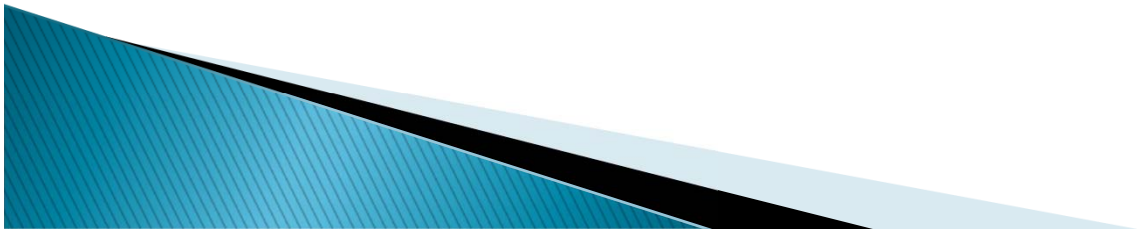


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```
17          <thead>
18             <!-- rowspans and colspans merge the specified -->
19             <!-- number of cells vertically or horizontally -->
20             <tr>
21                <!-- merge two rows -->
22                <th rowspan = "2">
23                   <img src = "camel.png" width = "205"
24                      height = "167" alt = "Picture of a camel">
25                </th>
26
27                <!-- merge four columns -->
28                <th colspan = "4">
29                   <strong>Camelid comparison</strong><br>
30                   Approximate as of 10/2011
31                </th>
32             </tr>
33             <tr>
34                <th># of humps</th>
35                <th>Indigenous region</th>
36                <th>Spits?</th>
37                <th>Produces wool?</th>
38             </tr>
39          </thead>
```

Fig. 2.13 | Complex HTML5 table. (Part 2 of 4.)

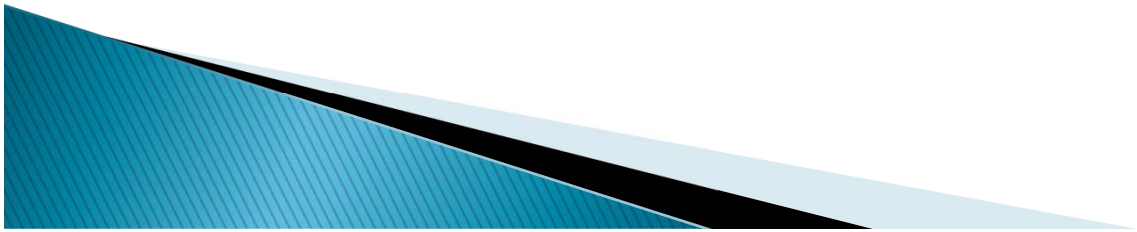




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```
40            <tbody>
41               <tr>
42                  <th>Camels (bactrian)</th>
43                  <td>2</td>
44                  <td>Africa/Asia</td>
45                  <td>Yes</td>
46                  <td>Yes</td>
47               </tr>
48               <tr>
49                  <th>Llamas</th>
50                  <td>1</td>
51                  <td>Andes Mountains</td>
52                  <td>Yes</td>
53                  <td>Yes</td>
54               </tr>
55            </tbody>
56         </table>
57      </body>
58   </html>
```

Fig. 2.13 | Complex HTML5 table. (Part 3 of 4.)





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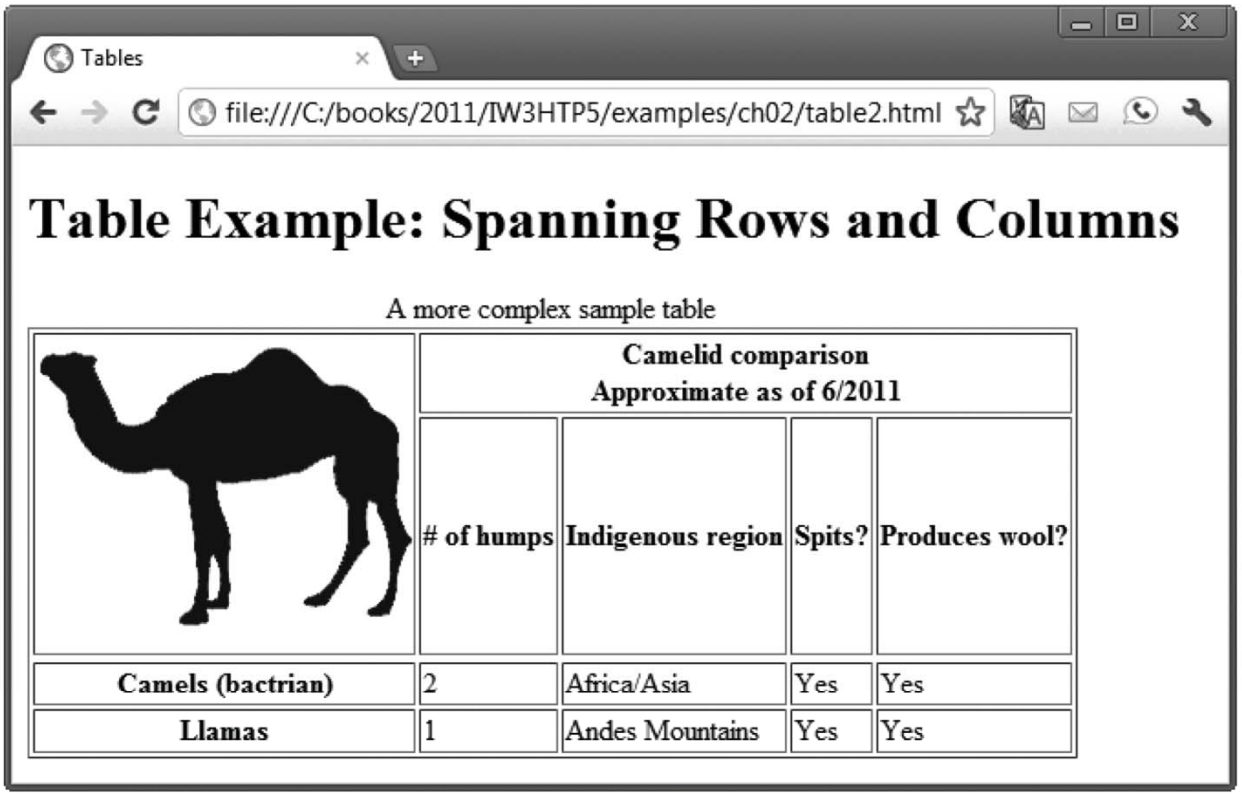


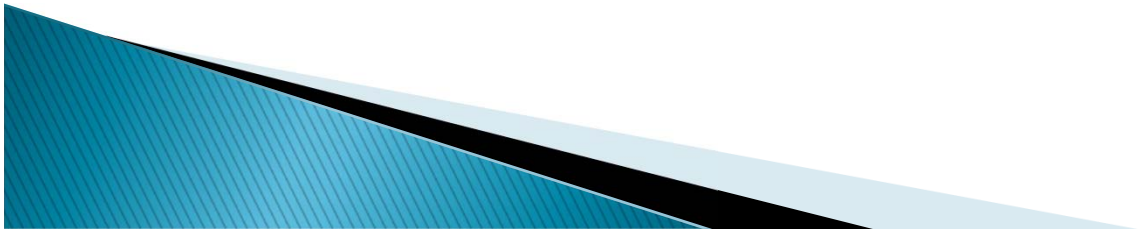
Fig. 2.13 | Complex HTML5 table. (Part 4 of 4.)

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2.11 Forms

- HTML5 provides **forms** for collecting information from users.
- Figure 2.14 is a simple form that sends data to the web server for processing.



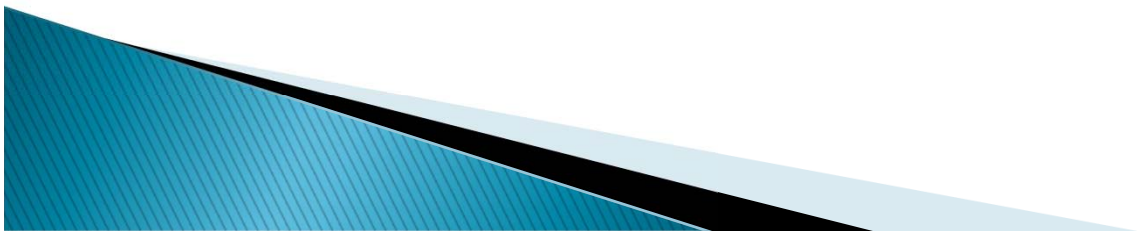


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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.14: form.html -->
4  <!-- Form with a text field and hidden fields. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>Forms</title>
9     </head>
10
11    <body>
12       <h1>Feedback Form</h1>
13
14       <p>Please fill out this form to help
15          us improve our site.</p>
16
17       <!-- this tag starts the the form, gives the -->
18       <!-- method of sending information and the -->
19       <!-- location of the form-processing script -->
20       <form method = "post" action = "http://www.deitel.com">
21          <!-- hidden inputs contain non-visual -->
22          <!-- information that will also be submitted -->
23          <input type = "hidden" name = "recipient"
24             value = "deitel@deitel.com">
```

Fig. 2.14 | Form with a text field and hidden fields. (Part 1 of 3.)





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```
25            <input type = "hidden" name = "subject"
26               value = "Feedback Form">
27            <input type = "hidden" name = "redirect"
28               value = "main.html">
29
30            <!-- <input type = "text"> inserts a text field -->
31            <p><label>Name:
32                  <input name = "name" type = "text" size = "25"
33                     maxlength = "30">
34               </label></p>
35
36            <p>
37               <!-- input types "submit" and "reset" insert -->
38               <!-- buttons for submitting and clearing the -->
39               <!-- form's contents, respectively -->
40               <input type = "submit" value = "Submit">
41               <input type = "reset" value = "Clear">
42            </p>
43         </form>
44      </body>
45   </html>
```

Fig. 2.14 | Form with a text field and hidden fields. (Part 2 of 3.)

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Fig. 2.14 | Form with a text field and hidden fields. (Part 3 of 3.)

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2.11 Forms (Cont.)

method Attribute of the form Element

- A form is defined by a `form` element
  - Attribute `method` specifies how the form's data is sent to the web server.
  - Using method = `"post"` appends form data to the browser request, which contains the protocol (HTTP) and the requested resource's URL.
  - The other possible value, method = `"get"`, appends the form data directly to the end of the URL of the script, where it's visible in the browser's Address field.
  - The `action` attribute of the form element specifies the script to which the form data will be sent



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2.11 Forms (Cont.)

`action` Attribute of the `form` Element

- The `action` attribute of the form element specifies the script to which the form data will be sent.
- Since we haven't introduced server-side programming yet, we set this attribute to `http://www.deitel.com` for now.
- **`input`** elements that specify data to provide to the script that processes the form (also called the **form handler**).
- An input's type is determined by its **`type` attribute**.



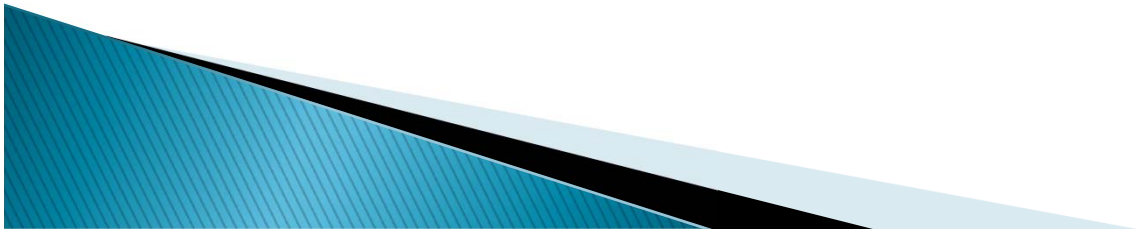
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2.11 Forms (Cont.)

Hidden Inputs

- Forms can contain visual and nonvisual components.
- Visual components include clickable buttons and other graphical user interface components with which users interact.
- Nonvisual components, called hidden inputs, store any data that you specify, such as e-mail addresses and HTML5 document file names that act as links.





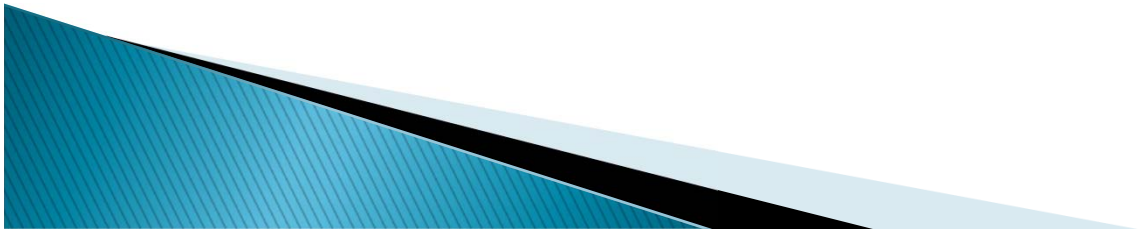
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2.11 Forms (Cont.)

`text input` Element

- The `text input` inserts a text field into the form, which allows the user to input data.
- The `label` element provides users with information about the input element's purpose
- The `size` attribute specifies the number of characters visible in the text field.
- Optional attribute `maxlength` limits the number of characters input into a text field.





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2.11 Forms (Cont.)

submit and reset input Elements

- The **submit** input element is a button.
  - When the submit button is pressed, the form's data is sent to the location specified in the form's action attribute.
- The **value** attribute sets the text displayed on the button.
- The **reset** input element allows a user to reset all form elements to their default values.



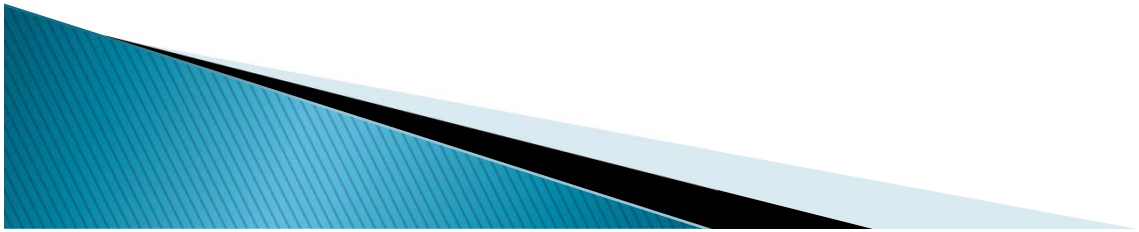
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2.11 Forms (Cont.)

Additional Form Elements

- Figure 2.15 contains a form that solicits user feedback about a website.
- The **`textarea`** element inserts a *multiline text area* into the form.
- The number of rows is specified with the **`rows`** attribute, and the number of columns (i.e., characters per line) with the **`cols`** attribute.
- Default text can be specified in other `input` types, such as text fields, by using the `value` attribute.



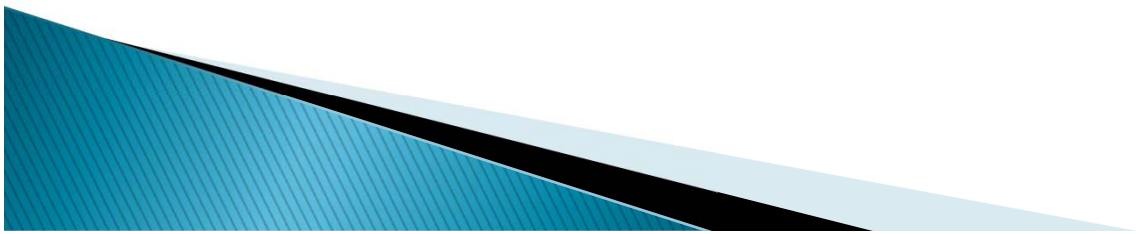


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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.15: form2.html -->
4  <!-- Form using a variety of components. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>More Forms</title>
9     </head>
10
11    <body>
12       <h1>Feedback Form</h1>
13       <p>Please fill out this form to help
14          us improve our site.</p>
15
16       <form method = "post" action = "http://www.deitel.com">
17
18          <input type = "hidden" name = "recipient"
19             value = "deitel@deitel.com">
20          <input type = "hidden" name = "subject"
21             value = "Feedback Form">
22          <input type = "hidden" name = "redirect"
23             value = "main.html">
24
```

Fig. 2.15 | Form using a variety of components. (Part 1 of 6.)



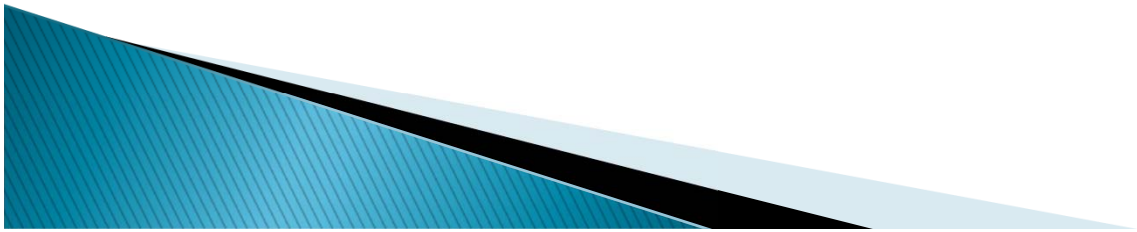


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```
25          <p><label>Name:
26             <input name = "name" type = "text" size = "25">
27          </label></p>
28
29          <!-- <textarea> creates a multiline textbox -->
30          <p><label>Comments:<br>
31             <textarea name = "comments"
32                rows = "4" cols = "36">Enter comments here.</textarea>
33          </label></p>
34
35          <!-- <input type = "password"> inserts a -->
36          <!-- textbox whose display is masked with -->
37          <!-- asterisk characters -->
38          <p><label>E-mail Address:
39             <input name = "email" type = "password"  size = "25">
40          </label></p>
41
42          <p>
43             <strong>Things you liked:</strong><br>
44
45             <label>Site design
46                <input name = "thingsliked" type = "checkbox"
47                   value = "Design"></label>
```

Fig. 2.15 | Form using a variety of components. (Part 2 of 6.)



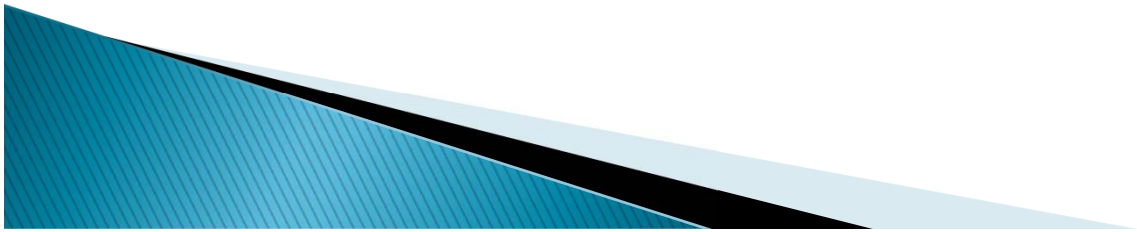


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```
48             <label>Links
49                <input name = "thingsliked" type = "checkbox"
50                   value = "Links"></label>
51             <label>Ease of use
52                <input name = "thingsliked" type = "checkbox"
53                   value = "Ease"></label>
54             <label>Images
55                <input name = "thingsliked" type = "checkbox"
56                   value = "Images"></label>
57             <label>Source code
58                <input name = "thingsliked" type = "checkbox"
59                   value = "Code"></label>
60          </p>
61
62          <!-- <input type = "radio"> creates a radio -->
63          <!-- button. The difference between radio buttons -->
64          <!-- and checkboxes is that only one radio button -->
65          <!-- in a group can be selected. -->
66          <p>
67             <strong>How did you get to our site?:</strong><br>
68
69             <label>Search engine
70                <input name = "howtosite" type = "radio"
71                   value = "search engine" checked></label>
```

Fig. 2.15 | Form using a variety of components. (Part 3 of 6.)





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```
72              <label>Links from another site
73                 <input name = "howtosite" type = "radio"
74                    value = "link"></label>
75              <label>Deitel.com Web site
76                 <input name = "howtosite" type = "radio"
77                    value = "deitel.com"></label>
78              <label>Reference in a book
79                 <input name = "howtosite" type = "radio"
80                    value = "book"></label>
81              <label>Other
82                 <input name = "howtosite" type = "radio"
83                    value = "other"></label>
84           </p>
85
86           <p>
87              <label>Rate our site:
88
89                 <!-- the <select> tag presents a drop-down -->
90                 <!-- list with choices indicated by the -->
91                 <!-- <option> tags -->
92              <select name = "rating">
93                 <option selected>Amazing</option>
94                 <option>10</option>
95                 <option>9</option>
96                 <option>8</option>
```

Fig. 2.15 | Form using a variety of components. (Part 4 of 6.)



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```
97                      <option>7</option>
98                      <option>6</option>
99                      <option>5</option>
100                     <option>4</option>
101                     <option>3</option>
102                     <option>2</option>
103                     <option>1</option>
104                     <option>Awful</option>
105                  </select>
106               </label>
107            </p>
108
109            <p>
110               <input type = "submit" value = "Submit">
111               <input type = "reset" value = "Clear">
112            </p>
113         </form>
114      </body>
115   </html>
```

Fig. 2.15 | Form using a variety of components. (Part 5 of 6.)

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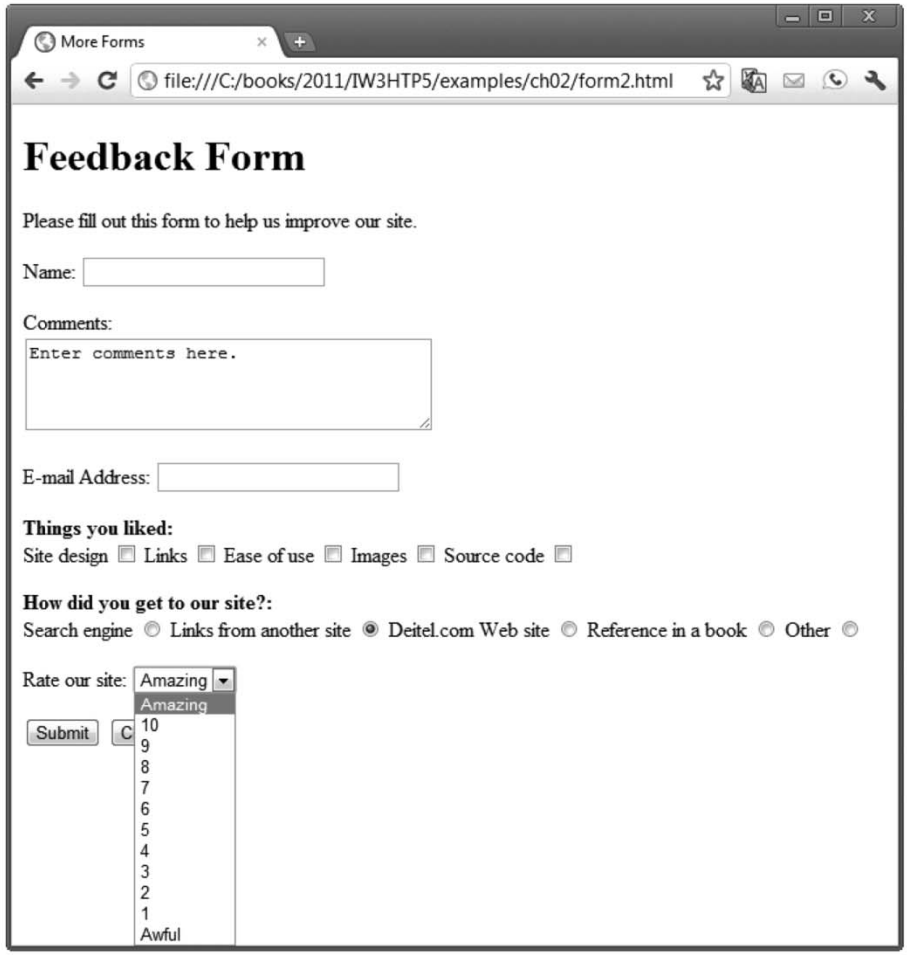


Fig. 2.15 | Form using a variety of components. (Part 6 of 6.)

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2.11 Forms (Cont.)

- The password input inserts a password box into a form.
  - Allows users to enter sensitive information, such as credit card numbers and passwords, by "masking" the information input with another character, usually asterisks.
  - The actual value input is sent to the web server, not the asterisks that mask the input.



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2.11 Forms (Cont.)

- The checkbox input element enables users to select and option.
  - When the checkbox is selected, a check mark appears in the checkbox . Otherwise, the checkbox is empty
  - checkboxes can be used individually and in groups. checkboxes that are part of the same group have the same name
- `radio` buttons are similar to checkboxes, except that only one radio button in a group can be selected at any time.
  - All radio buttons in a group have the same name attribute but different value attributes.
- The `select` input provides a drop-down list of items.
  - The `name` attribute identifies the drop-down list.
  - The `option` element adds items to the drop-down list.



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Common Programming Error 2.1

When your `form` has several checkboxes with the same `name`, make sure that they have different `values`, or the web server scripts will not be able to distinguish them.

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Common Programming Error 2.2

Not setting the `name` attributes of the `radio` buttons in a group to the same name is a logic error because it lets the user select all of the `radio` buttons at the same time.

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2.12 Internal Linking

- The a tag can be used to link to another section of the same document by specifying the element's `id` as the link's `href`.
- To link internally to an element with its `id` attribute set, use the syntax *#id*.



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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.16: internal.html -->
4  <!-- Internal Linking -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>Internal Links</title>
9     </head>
10
11    <body>
12       <!-- id attribute creates an internal hyperlink destination -->
13       <h1 id = "features">The Best Features of the Internet</h1>
14
15       <!-- an internal link's address is "#id" -->
16       <p><a href = "#bugs">Go to <em>Favorite Bugs</em></a></p>
17
18       <ul>
19          <li>You can meet people from countries
20             around the world.</li>
21          <li>You have access to new media as it becomes public:
22             <ul>
23                <li>New games</li>
```

Fig. 2.16 | Internal hyperlinks to make pages more navigable. (Part 1 of 5.)

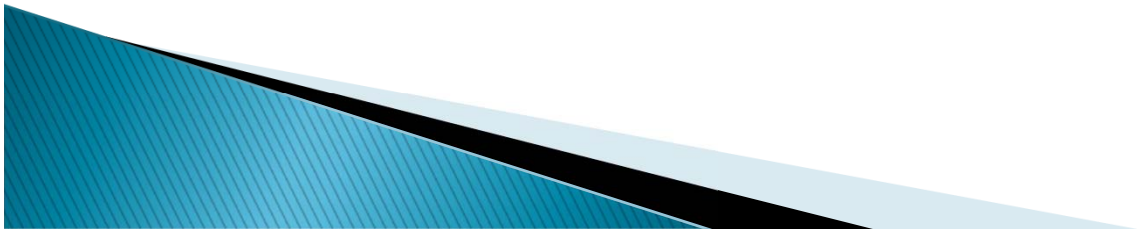


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```
24                <li>New applications
25                   <ul>
26                      <li>For Business</li>
27                      <li>For Pleasure</li>
28                   </ul>
29                </li>
30
31                <li>Around the clock news</li>
32                <li>Search Engines</li>
33                <li>Shopping</li>
34                <li>Programming
35                   <ul>
36                      <li>HTML5</li>
37                      <li>Java</li>
38                      <li>Dynamic HTML</li>
39                      <li>Scripts</li>
40                      <li>New languages</li>
41                   </ul>
42                </li>
43             </ul>
44          </li>
45
```

Fig. 2.16 | Internal hyperlinks to make pages more navigable. (Part 2 of 5.)



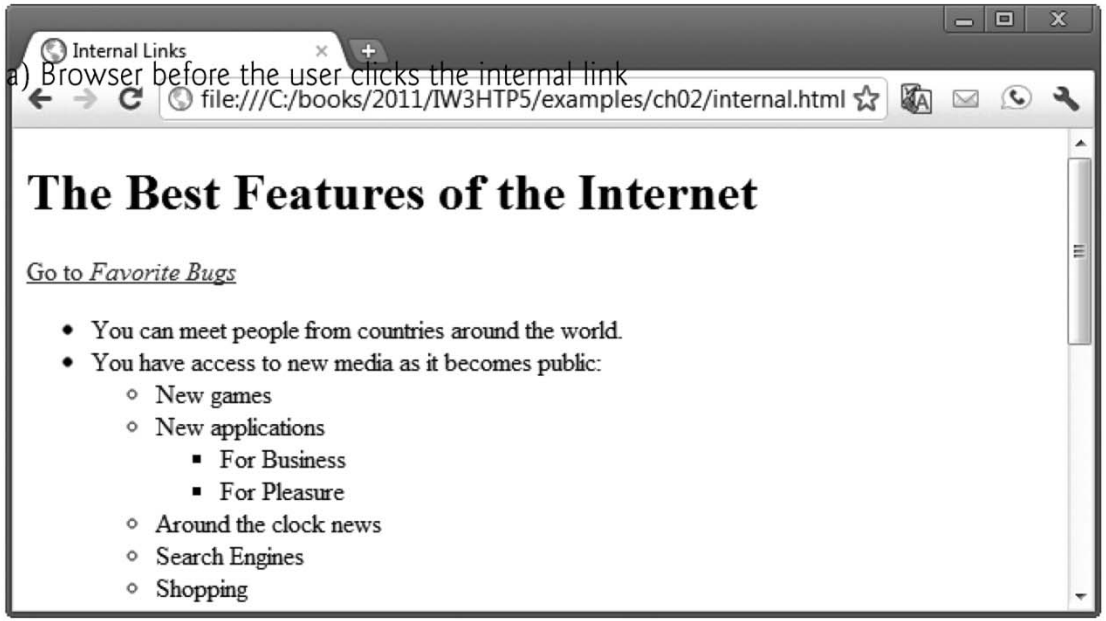


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```
46            <li>Links</li>
47            <li>Keeping in touch with old friends</li>
48            <li>It is the technology of the future!</li>
49         </ul>
50
51         <!-- id attribute creates an internal hyperlink destination -->
52         <h1 id = "bugs">My 3 Favorite Bugs</h1>
53         <p>
54            <!-- internal hyperlink to features -->
55            <a href = "#features">Go to <em>Favorite Features</em></a>
56         </p>
57         <ol>
58            <li>Fire Fly</li>
59            <li>Gal Ant</li>
60            <li>Roman Tic</li>
61         </ol>
62      </body>
63   </html>
```

Fig. 2.16 | Internal hyperlinks to make pages more navigable. (Part 3 of 5.)

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a) Browser before the user clicks the internal link

Fig. 2.16 | Internal hyperlinks to make pages more navigable. (Part 4 of 5.)

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Fig. 2.16 | Internal hyperlinks to make pages more navigable. (Part 5 of 5.)

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2.13 meta Elements

- One way that search engines catalog pages is by reading the meta element's contents.
  - The name attribute identifies the type of meta element
  - The content attribute
    - Of a keywords meta element: provides search engines with a list of words that describe a page, which are compared with words in search requests
    - Of a description meta element: provides a three- to four-line description of a site in sentence form, used by search engines to catalog your site. This text is sometimes displayed as part of the search result

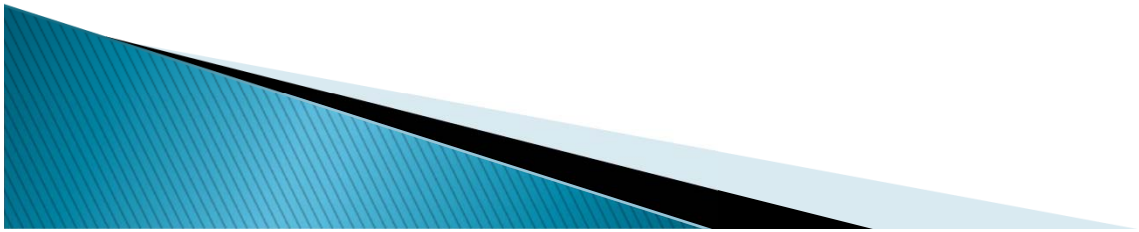


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```
1  <!DOCTYPE html>
2
3  <!-- Fig. 2.17: meta.html -->
4  <!-- meta elements provide keywords and a description of a page. -->
5  <html>
6     <head>
7        <meta charset = "utf-8">
8        <title>Welcome</title>
9
10       <!-- <meta> tags provide search engines with -->
11       <!-- information used to catalog a site      -->
12       <meta name = "keywords" content = "web page, design,
13          HTML5, tutorial, personal, help, index, form,
14          contact, feedback, list, links, deitel">
15       <meta name = "description" content = "This website will
16          help you learn the basics of HTML5 and web page design
17          through the use of interactive examples and
18          instruction.">
19    </head>
```

Fig. 2.17 | meta elements provide keywords and a description of a page. (Part 1 of 3.)



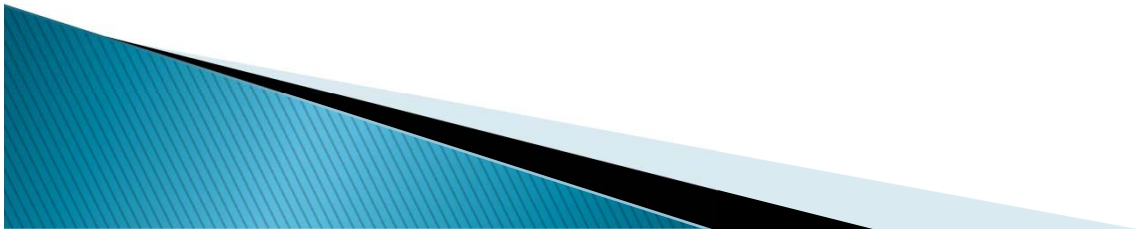


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```
20      <body>
21         <h1>Welcome to Our Website!</h1>
22
23         <p>We have designed this site to teach about the wonders
24         of <strong><em>HTML5</em></strong>. <em>HTML5</em> is
25         better equipped than <em>HTML</em> to represent complex
26         data on the Internet. <em>HTML5</em> takes advantage of
27         XML's strict syntax to ensure well-formedness. Soon you
28         will know about many of the great features of
29         <em>HTML5.</em></p>
30
31         <p>Have Fun With the Site!</p>
32      </body>
33   </html>
```

Fig. 2.17 | meta elements provide keywords and a description of a page. (Part 2 of 3.)





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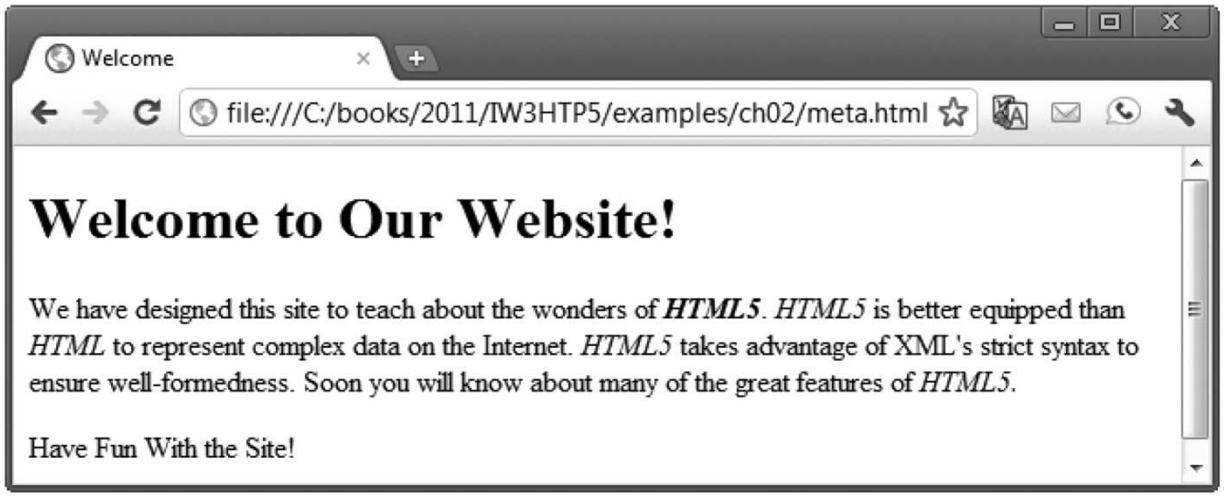


Fig. 2.17 | `meta` elements provide keywords and a description of a page. (Part 3 of 3.)

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Software Engineering Observation 2.2

`meta` elements are not visible to users. They must be placed inside the `head` section of your HTML5 document; otherwise they will not be read by search engines.

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